

# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of: Notification No. 2023-9 of the Ministry of Employment and Labour Standards for Classification, Labelling of Chemical Substances and Material Safety Data Sheets

**Issuing Date** 28-Jan-2026

**Revision date** 29-Jan-2026

**Revision Number** 1.02

## 1. Identification

### A. Product identifier

**Product Name** Anti-MMP2 antibody [EPR1184]

**Product Code(s)** AB92536

### B. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Laboratory chemicals, For research use only. Not for use in diagnostic procedures

**Restrictions on use**

### C. Supplier's details

#### **Importer**

DAWINBIO Inc.  
601, ITECO, 150,  
Jojeong-daero,  
Hanam-city,  
Gyeonggi-do, 12930,  
South Korea  
Tel : +82-1588-9741  
Fax : +82-31-728-3200

#### **Supplier**

Abcam Singapore Pte Ltd.  
11 North Buona Vista Drive,  
#16-08, The Metropolis Tower Two,  
Singapore 138589  
Office Tel: +65 6734 9252  
Office Fax: +65 6722 0608  
Toll free number: 800 321 1284

For further information, please contact

**E-mail address** sg.technical@abcam.com, sds@abcam.com

**Emergency telephone number** +82 2 3479 8401

## 2. Hazard(s) identification

### A. Classification of the substance or mixture

Not a hazardous substance or mixture according to the Globally Harmonized System (GHS)  
Not classified

### B. GHS Label elements, including precautionary statements

#### **Hazard symbols**

Not applicable

#### **Hazard statements**

No hazard statements required.

#### **Precautionary Statements - Prevention**

P280 - Wear protective gloves, protective clothing, eye protection and face protection

**C. Other hazards which do not result in classification**

No information available.

**3. Composition/information on ingredients****Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health.

| Chemical name                  | Common name and synonyms | CAS No. | Weight-% | Korean Existing Chemicals Inventory |
|--------------------------------|--------------------------|---------|----------|-------------------------------------|
| Glycerol (glycerin, glycerine) | No information available | 56-81-5 | 40 - <45 | KE-29297                            |

**4. First-aid measures****A. In case of eye contact****In case of eye contact**

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.  
Consult a physician.

**B. In case of skin contact**

Wash skin with soap and water.

**C. In case of inhalation**

Remove to fresh air.

**D. In case of ingestion**

Rinse mouth.

**E. Indication of immediate medical attention and special treatment needed, if necessary****Note to physicians**

Treat symptomatically.

**Symptoms**

See Section 2.2 for more information.

**Effects of Exposure**

No information available.

**5. Fire-fighting measures****A. Suitable (and unsuitable) extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire**

CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media**

Do not scatter spilled material with high pressure water streams.

**B. Specific hazards arising from the chemical**

No information available.

**C. Special Protective Equipment for Firefighters**

**Special Protective Equipment for Firefighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.  
Use personal protection equipment.

**6. Accidental release measures****A. Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**For emergency responders** Use personal protection recommended in Section 8.

**B. Environmental precautions**

See Section 12 for additional Ecological Information.

**C. Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers. Clean contaminated surface thoroughly.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****A. Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**B. Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**8. Exposure controls/personal protection****A. Control Parameters****Occupational exposure limits**

| Chemical name                  | OEL                              | PEL | ACGIH TLV |
|--------------------------------|----------------------------------|-----|-----------|
| Glycerol (glycerin, glycerine) | TWA: 10 mg/m <sup>3</sup> ; mist | -   | -         |

**Note** See section 16 for terms and abbreviations

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**B. Appropriate engineering controls**

**Engineering controls** Showers

|                                         |                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | Eyewash stations<br>Ventilation systems.                                                                                                                                         |
| <b>Environmental exposure controls</b>  | No information available.                                                                                                                                                        |
| <b>C. Personal protective equipment</b> |                                                                                                                                                                                  |
| <b>Respiratory protection</b>           | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.         |
| <b>Eye protection</b>                   | Use suitable eye protection.                                                                                                                                                     |
| <b>Hand protection</b>                  | Wear suitable gloves. Ensure that the breakthrough time of the glove material is not exceeded. Refer to glove supplier for information on breakthrough time for specific gloves. |
| <b>Body protection</b>                  | Wear suitable protective clothing.                                                                                                                                               |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                          |                          |
|--------------------------|--------------------------|
| <b>A. Appearance</b>     | aqueous solution         |
| <b>Physical State</b>    | Liquid                   |
| <b>Color</b>             | colorless                |
| <b>B. Odor</b>           | Odorless                 |
| <b>C. Odor threshold</b> | No information available |

| <u>Property</u>                                        | <u>Values</u>            | <u>Remarks • Method</u> |
|--------------------------------------------------------|--------------------------|-------------------------|
| <b>D. pH</b>                                           | 7.2 - 7.4                |                         |
| <b>E. Melting point / freezing point</b>               | No information available | None known              |
| <b>F. Initial boiling point and boiling range</b>      | No information available | None known              |
| <b>G. Flash point</b>                                  | No information available | None known              |
| <b>H. Evaporation rate</b>                             | No information available | None known              |
| <b>I. Flammability</b>                                 | No information available | None known              |
| <b>J. Upper/lower flammability or explosive limits</b> |                          |                         |
| Upper flammability or explosive limits                 | No information available |                         |
| Lower flammability or explosive limits                 | No information available |                         |
| <b>K. Vapor pressure</b>                               | No information available | None known              |
| <b>L. Solubility(ies)</b>                              |                          |                         |
| Water solubility                                       |                          | None known              |
| Solubility in other solvents                           | No information available | None known              |
| <b>M. Relative vapor density</b>                       | No information available | None known              |
| <b>N. Specific Gravity</b>                             | No information available | None known              |
| Bulk density                                           | No information available |                         |
| Liquid Density                                         | No information available |                         |
| <b>O. Partition coefficient: n-octanol/water</b>       | No information available | None known              |
| <b>P. Autoignition Point</b>                           | No information available | None known              |
| <b>Q. Decomposition temperature</b>                    |                          | None known              |
| SADT (°C)                                              | No information available | None known              |
| <b>R. Viscosity</b>                                    |                          |                         |
| Kinematic viscosity                                    | No information available | None known              |

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|                                 |                          |            |
|---------------------------------|--------------------------|------------|
| <b>Dynamic viscosity</b>        | No information available | None known |
| <b>S. Molecular weight</b>      | No information available |            |
| <b><u>Other information</u></b> | No information available |            |

## **10. Stability and reactivity**

### **A. Chemical stability and possibility of hazardous reactions**

**Stability** Stable under normal conditions.

**Possibility of hazardous reactions** None under normal processing.

#### **Explosion data**

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### **B. Conditions to avoid**

None known based on information supplied.

### **C. Incompatible materials**

None known based on information supplied.

### **D. Hazardous decomposition products**

None known based on information supplied.

## **11. Toxicological information**

### **A. Information on the likely routes of exposure**

#### **Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Symptoms** See Section 2.2 for more information.

### **B. Health hazards information**

**Acute toxicity** No information available.

#### **Numerical measures of toxicity**

**The following ATE values have been calculated for the mixture**

**ATEmix (oral)** 99,999.00 mg/kg

**ATEmix (dermal)** 99,999.00 mg/kg

**ATEmix (inhalation-gas)** 99,999.00 ppm

ATEmix (inhalation-vapor) 99,999.000 mg/L  
 ATEmix (inhalation-dust/mist) 99,999.00 mg/L

**Component Information**

| Chemical name                  | Oral LD50             | Dermal LD50          | Inhalation LC50         |
|--------------------------------|-----------------------|----------------------|-------------------------|
| Glycerol (glycerin, glycerine) | = 27200 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 5.85 mg/L ( Rat ) 4 h |

**Skin corrosion/irritation** No information available.

**Serious eye damage/irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Carcinogenicity** No information available.

**Germ cell mutagenicity** No information available.

**Reproductive toxicity** No information available.

**Specific target organ toxicity (STOT)** No information available.  
 - single exposure

**Specific target organ toxicity (STOT)** No information available.  
 - repeated exposure

**Target organ effects** No information available.

**Aspiration hazard** No information available.

**12. Ecological information****A. Ecotoxicity**

Based on available data, the classification criteria are not met.

**Aquatic ecotoxicity****Component Information**

| Chemical name                  | Fish                                         | Crustacea | Algae/aquatic plants | Toxicity to microorganisms |
|--------------------------------|----------------------------------------------|-----------|----------------------|----------------------------|
| Glycerol (glycerin, glycerine) | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | -         | -                    | -                          |

**B. Persistence and degradability**

No information available.

**C. Bioaccumulative potential**

| Chemical name                  | Partition coefficient | Bioconcentration factor (BCF) | Trophic magnification factor (TMF) |
|--------------------------------|-----------------------|-------------------------------|------------------------------------|
| Glycerol (glycerin, glycerine) | -1.75                 | -                             | -                                  |

**D. Mobility in soil**

No information available.

**E. Other adverse effects**

No information available.

**13. Disposal considerations****A. Disposal methods****Waste from residues/unused products**

Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**B. Disposal considerations****Contaminated packaging**

Do not reuse empty containers.

**14. Transport information**

|                                        |               |
|----------------------------------------|---------------|
| <b>A. UN number</b>                    | Not regulated |
| <b>B. UN proper shipping name</b>      | Not regulated |
| <b>C. Transport hazard class(es)</b>   | Not regulated |
| <b>D. Packing group</b>                | Not regulated |
| <b>E. Marine pollutant</b>             | Not regulated |
| <b>F. Special precautions for user</b> | Not regulated |

**15. Regulatory information****A. Industrial Safety and Health Law** Not applicable**Prohibited substance** Not applicable**Substances Requiring Permission** Not applicable**Harmful substances subject to control** Not applicable**Harmful agents subject to work environment monitoring** Not applicable**Harmful agents subject to workers requiring health examination** Not applicable**Harmful or dangerous substances subject to submission of process safety reports** Not applicable  
**Control parameters**

| Chemical name | OEL | PEL |
|---------------|-----|-----|
|---------------|-----|-----|

|                                |                                  |   |
|--------------------------------|----------------------------------|---|
| Glycerol (glycerin, glycerine) | TWA: 10 mg/m <sup>3</sup> ; mist | - |
|--------------------------------|----------------------------------|---|

**B. Chemicals Control Act** Not applicable

**Hazardous chemical substances** Not applicable

**C. Act on Registration, Evaluation, etc. of Chemicals (K-REACH)** Applicable

| Chemical name                  | Existing substances subject to registration | Existing substances not likely to be subject to registration | Existing substances known to be of very low risk | Substances with unknown hazards |
|--------------------------------|---------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|---------------------------------|
| Glycerol (glycerin, glycerine) | Not applicable                              | Applicable                                                   | Not applicable                                   | Not applicable                  |

| Chemical name                  | Korean Existing Chemicals Inventory | Priority control substances |
|--------------------------------|-------------------------------------|-----------------------------|
| Glycerol (glycerin, glycerine) | KE-29297                            | Not applicable              |
| Sodium dihydrogen phosphate    | KE-31577                            | Not applicable              |
| Sodium chloride                | KE-31387                            | Not applicable              |
| BSA                            | KE-05-0011                          | Not applicable              |
| Sodium azide                   | KE-31357                            | Not applicable              |

**D. Safety Control of Dangerous Substances Act** Not applicable

**E. Wastes Management** Dispose of waste in accordance with environmental legislation.

**F. Other Regulations** No information available

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

|               |                 |
|---------------|-----------------|
| TSCA          | Does not comply |
| DSL/NDSL      | Does not comply |
| EINECS/ELINCS | Does not comply |
| ENCS          | Does not comply |
| IECSC         | Does not comply |
| KECL          | Does not comply |
| PICCS         | Does not comply |
| AIIC          | Does not comply |
| NZIoC         | Complies        |
| TCSI          | Does not comply |

| Chemical name                  | TSCA   | DSL | NDSL | EINECS | ELINCS |
|--------------------------------|--------|-----|------|--------|--------|
| Glycerol (glycerin, glycerine) | ACTIVE | X   | -    | X      | -      |

| Chemical name                  | ENCS | IECSC | KECL | PICCS | AIIC | NZIoC | TCSI |
|--------------------------------|------|-------|------|-------|------|-------|------|
| Glycerol (glycerin, glycerine) | X    | X     | X    | X     | X    | -     | X    |



**Legend:** X - Listed**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances**ENCS** - Japan Existing and New Chemical Substances**IECSC** - China Inventory of Existing Chemical Substances**KECL** - Korean Existing Chemicals Inventory**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**AIIC** - Australian Inventory of Industrial Chemicals**NZIoC** - New Zealand Inventory of Chemicals**TCSI** - Taiwan Chemical Substance Inventory**16. Other information****A. Information source and references**

**Prepared By** No information available.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

|         |                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------|
| ACGIH   | American Conference of Governmental Industrial Hygienists                                           |
| ADN     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe)     |
| ADR     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)                 |
| AIIC    | Australian Inventory of Industrial Chemicals                                                        |
| ATE     | Acute Toxicity Estimate                                                                             |
| ASTM    | American Society for the Testing of Materials                                                       |
| bar     | Biological Reference Values for Chemical Compounds in the Work Area                                 |
| BAT     | Biological tolerance values for occupational exposure                                               |
| BEL     | Biological exposure limits                                                                          |
| bw      | Body weight                                                                                         |
| Ceiling | Maximum limit value                                                                                 |
| CMR     | Carcinogen, Mutagen or Reproductive Toxicant                                                        |
| DOT     | Department of Transportation (United States)                                                        |
| DSL     | Domestic Substances List (Canada)                                                                   |
| EmS     | Emergency Schedule                                                                                  |
| ENCS    | Existing and New Chemical Substances (Japan)                                                        |
| EPA     | U.S. Environmental Protection Agency                                                                |
| GHS     | Globally Harmonized System                                                                          |
| IARC    | International Agency for Research on Cancer                                                         |
| IATA    | International Air Transport Association                                                             |
| IBC     | International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk |
| ICAO    | International Civil Aviation Organization                                                           |
| IECSC   | Inventory of Existing Chemical Substances in China                                                  |
| IMDG    | International Maritime Dangerous Goods                                                              |
| IMO     | International Maritime Organization                                                                 |
| ISO     | International Organization for Standardization                                                      |
| KECI    | Korean Existing Chemicals Inventory                                                                 |
| LC50    | Lethal Concentration to 50% of a test population                                                    |
| LD50    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                        |
| MARPOL  | International Convention for the Prevention of Pollution from Ships                                 |
| MSDS    | Material Safety Data Sheet                                                                          |
| n.o.s.  | Not Otherwise Specified                                                                             |
| NOAEC   | No Observed Adverse Effect Concentration                                                            |

|         |                                                                                     |
|---------|-------------------------------------------------------------------------------------|
| NOAEL   | No Observed Adverse Effect Level                                                    |
| NOELR   | No Observable Effect Loading Rate                                                   |
| NZIoC   | New Zealand Inventory of Chemicals                                                  |
| OECD    | Organization for Economic Cooperation and Development                               |
| OEL     | Occupational exposure limits                                                        |
| PBT     | Persistent, Bioaccumulative and Toxic substance                                     |
| PICCS   | Philippines Inventory of Chemicals and Chemical Substances                          |
| PMT     | Persistent, Mobile and Toxic                                                        |
| PPE     | Personal protective equipment                                                       |
| QSAR    | Quantitative Structure Activity Relationship                                        |
| RID     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| SADT    | Self-Accelerating Decomposition Temperature                                         |
| SAR     | Structure-activity relationship                                                     |
| SDS     | Safety Data Sheet                                                                   |
| SL      | Surface Limit                                                                       |
| STEL    | Short Term Exposure Limit                                                           |
| STOT RE | Specific target organ toxicity - Repeated exposure                                  |
| STOT SE | Specific target organ toxicity - Single exposure                                    |
| TCSI    | Taiwan Chemical Substance Inventory                                                 |
| TDG     | Transport of Dangerous Goods (Canada)                                               |
| TSCA    | Toxic Substances Control Act (United States)                                        |
| TWA     | Time-Weighted Average                                                               |
| UN      | United Nations                                                                      |
| VOC     | Volatile organic compounds                                                          |
| vPvB    | Very Persistent and Very Bioaccumulative                                            |
| vPvM    | Very Persistent and Very Mobile                                                     |
| As      | Allergenic substance                                                                |
| DS      | Dermal Sensitizer                                                                   |
| Ot      | Ototoxicant                                                                         |
| pOt     | Ototoxicant - potential to cause hearing disorders                                  |
| PS      | Photosensitizer                                                                     |
| RS      | Respiratory Sensitizer                                                              |
| S       | Sensitizer                                                                          |
| poS     | Sensitizer - capable of causing occupational asthma                                 |
| Sa      | Simple asphyxiant                                                                   |
| Sd      | Skin designation                                                                    |
| pSd     | Skin designation - potential for cutaneous absorption                               |
| Sdv     | Skin designation - vacated                                                          |
| Sk      | Skin notation                                                                       |
| dSk     | Skin notation - danger of cutaneous absorption                                      |
| pSk     | Skin notation - potential for cutaneous absorption                                  |

**Key literature references and sources for data used to compile the SDS**

U.S. Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 U.S. Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)

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National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
International Organization for Economic Co-operation and Development (OECD) Environment, Health, and Safety Publications  
International Organization for Economic Co-operation and Development (OECD) High Production Volume Chemicals Program  
International Organization for Economic Co-operation and Development (OECD) Screening Information Data Set  
United Nations World Health Organization (WHO)

**B. Issuing Date** 28-Jan-2026

**C. Revision number and date**

|                        |             |
|------------------------|-------------|
| <b>Revision Number</b> | 1.02        |
| <b>Revision date</b>   | 29-Jan-2026 |

**D. Other**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**